



ST BENEDICT'S

SUBJECT	CHEMISTRY MID-YEAR EXAM
GRADE	11
EXAMINER	Mr Hilder
MODERATOR	Mr McCoy
DATE	22 July 2022
MARKS	165
DURATION	150 minutes

MARK
TOTAL

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LEARNER NAME: **MEMO - FINAL**

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 19 pages and a Data Sheet of 2 pages, please detach the Data Sheet.
Please check that your question paper is complete.
2. Read the questions carefully.
3. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment. Remember to write your name and surname in the space provided above.
4. Use the data and formulae sheet whenever necessary.
5. Show your working in all calculations.
6. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
7. Answers must be expressed in decimal format, not left as proper fractions.
8. Where appropriate, express answers to TWO decimal places.
9. It is in your own interest to write legibly and to present your work neatly.
10. Blank pages are included at the end of the paper (page 20). If you run out of space for a question, use this page. Clearly indicate the question number of your answer should you use this extra space.

FOR OFFICE USE ONLY: MARKER TO ENTER MARKS

Question	1	2	3	4	5	6	7	8	9	10	Total
Mark											
Moderated Mark											
Question Total	16	19	12	11	26	17	12	13	19	20	165

Comment:

QUESTION 1 -MULTIPLE CHOICE**16 MARKS**

Answer these questions on the multiple-choice answer sheet below. Make a cross (X) in the box corresponding to the letter that you consider to be correct.

A	B	☒	D
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Here the option C has been marked as an example

1.1	A	B	C	D✓✓✓
1.2	A	B✓✓✓	C	D
1.3	A	B	C✓✓✓	D
1.4	A	B✓✓✓	C	D
1.5	A✓✓✓	B	C	D
1.6	A	B	C✓✓✓	D
1.7	A	B✓✓✓	C	D
1.8	A	B	C	D✓✓✓

1.1 The chemical formula for magnesium phosphate is: (2)

- A $\text{Mg}(\text{PO}_4)_2$
- B $\text{Mg}_2(\text{PO}_4)_3$
- C MgPO_4
- D $\text{Mg}_3(\text{PO}_4)_2$

1.2 Which is the correct exponent for the prefix *micro*. (2)

- A 10^{-9}
- B 10^{-6}
- C 10^{-3}
- D 10^{-2}

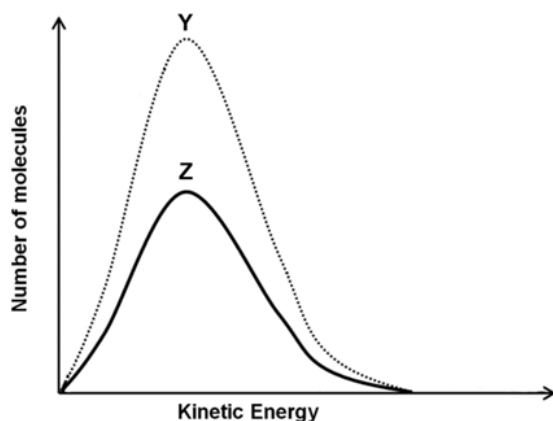
1.3 The number of neutrons in an atom of $^{14}_6\text{C}$ is... (2)

- A 6
- B 7
- C 8
- D 14

1.4 Which one of the following has a tetrahedral shape? (2)

- A H_3O^+
- B NH_4^+
- C CO_2
- D AlCl_3

- 1.5 The Maxwell-Boltzmann curve labelled Z shows the distribution of molecular energies of fluorine gas (F₂).



Which of the following correctly describes the change resulting in curve Y? (2)

- A An increase in concentration of fluorine gas only.
- B An increase in temperature of fluorine gas only.
- C An increase in concentration and increase in temperature of fluorine gas.
- D The addition of a catalyst.

- 1.6 The half reaction $\text{Sn}^{4+} + 2\text{e}^- \rightarrow \text{Sn}^{2+}$ shows the... (2)

- A oxidation of a metal ion.
- B oxidation of a non-metal ion.
- C reduction of a metal ion.
- D reduction of a non-metal ion.

- 1.7 Which sample of 20 g of calcium, oxygen gas, or sodium fluoride has the greatest number of moles? (2)

- A calcium
- B oxygen
- C sodium fluoride
- D All of the above have the same number of moles in 20 g

- 1.8 How many atoms are present in 6,4 moles of sulphuric acid, H₂SO₄. (2)

- A $1,16 \times 10^{24}$
- B $3,84 \times 10^{24}$
- C $4,21 \times 10^{24}$
- D $2,70 \times 10^{25}$

QUESTION 2

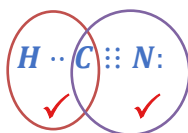
19 MARKS

Hydrogen cyanide (HCN) is a very toxic compound used in manufacturing of plastics, mining gold, and as a poison.

- 2.1 Define the term *chemical bond*. (2)

a bond which occurs between atoms within molecules ✓✓

- 2.2 Draw the Lewis structures for HCN. (2)



- 2.3 What is the shape of the HCN molecule? (1)

Linear ✓

- 2.4 Calculate the electronegativity difference for the CN bond. (2)

$$END = N - C$$

$$END = 3 - 2,5$$

$$END = 0,5$$

- 2.5 Is HCN a **POLAR** or **NON-POLAR** molecule? Circle the correct answer. (1)

The table below indicates the values of the bond length and bond energy of the different bonds of HCN.

BOND	BOND LENGTH (nm)	BOND ENERGY (kJ.mol ⁻¹)
CH	0,109	413
CN	0,116	890

- 2.6 Explain why the bond energy of the CN bond is more than the bond energy of the CH bond. (2)

CN ($C \equiv N$) has a triple bond and CH ($C - H$) only has a single bond. ✓

Triple bonds require more energy to break, as END are very similar. ✓

|OR| 6 e⁻ shared between C and N ✓, 2 e⁻ between C and H ✓ (c.o.e - 2.2)

- 2.7 Explain the difference between the bond length of the CH bond and the bond length of the CN bond (2)

Hydrogen is a smaller atom compared to nitrogen. ✓

H has 1 electron shell N has 2. Bond length is distance between nuclei.

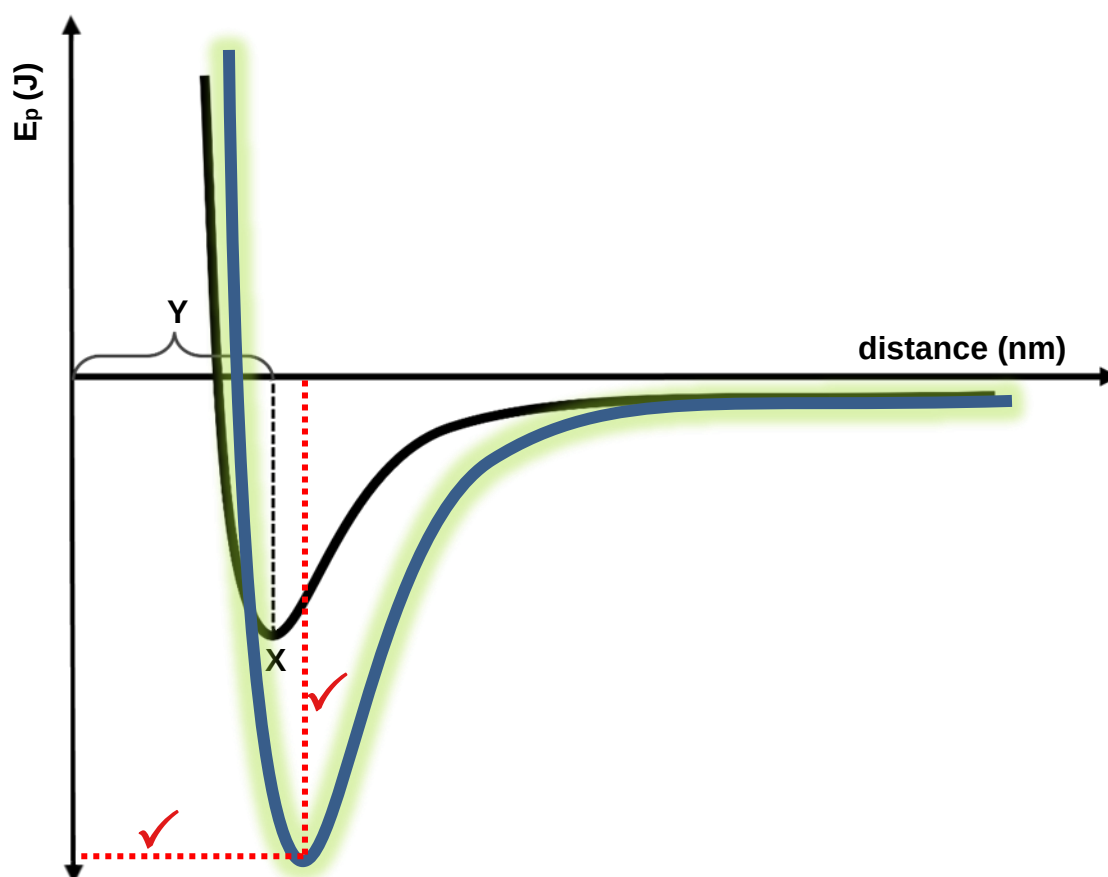
So C and can closer to H than N. ✓ (EN is basically irrelevant)

2.8 Will HCN be soluble in water? Circle only **YES** or **NO**. (1)

2.9 Explain your answer to QUESTION 2.8. (2)

HCN is a polar molecule due to unequal electron cloud density (ΔEN) ✓
Water is also polar, so HCN will be able to dissolve in the liquid. ✓
(like dissolves like)

2.10 The diagram below shows the energy change that takes place when carbon and hydrogen atoms move towards each other. The x-axis represents the distance between the nuclei of the two atoms.



2.10.1 What do each of the labelled points on the graph represent? (2)

X	Bond energy ✓
Y	Bond length ✓ (aon)

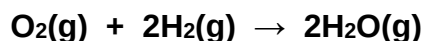
2.10.2 Draw another curve on the graph that represents carbon and nitrogen moving towards one another. (2)

marks awarded for:
 -> vertex lower on y-axis and
 -> vertex further right on x-axis.

QUESTION 3

12 MARKS

3.1 The reaction below is used in hydrogen fuel cells.



The boiling points of the substances in the reaction are as follows:

SUBSTANCE	BOILING POINT(°C)
H ₂	-252,9
O ₂	-195,8
H ₂ O	100,0

3.1.1 Refer to the intermolecular forces and explain the difference in boiling point between H₂O and O₂. (4)

O ₂ is a non-polar so have London forces. ✓
H ₂ O is polar and have Hydrogen Bonds. ✓
Takes more energy to overcome Hydrogen bonds compared to London. ✓
So water has the higher BP. ✓

3.1.2 Explain why H₂ will evaporate faster than O₂. Refer to the type and relative strength of the intermolecular forces. (4)

Both H ₂ and O ₂ are non-polar molecules and have London Forces. ✓
O ₂ is a larger molecule and have a larger e ⁻ cloud. ✓
This requires more energy to separate the molecules. ✓
So H ₂ will evaporate faster. ✓

3.2 When hydrochloric acid is added to water one of the products is the hydronium ion.

3.2.1 What chemical properties are required for a dative covalent bond to form. (2)

Empty e ⁻ orbital on one atom ✓
Lone pair of e ⁻ on the other ✓

3.2.2 Draw a Lewis diagram of the hydronium ion. (2)

✓ (3 H's and an O) ✓ [square bracket and dative symbol]⁺

QUESTION 4

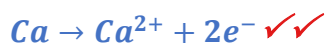
11 MARKS

4.1 Consider the reaction between calcium metal and chlorine to form calcium chloride.

4.1.1 Define the term *reducing agent*. (1)

A substance that donates electrons ✓

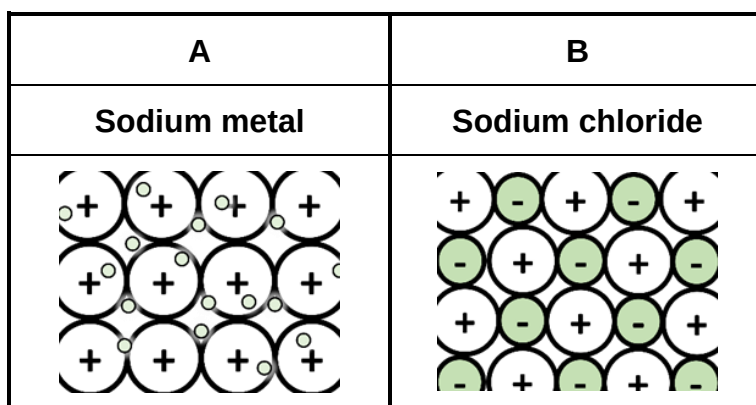
4.1.2 Write down the oxidation half reaction for the formation of calcium chloride. (2)



4.1.3 Give the net ionic equation for this reaction. (3)



4.2 Consider the particle diagram below (A and B) representing sodium metal and sodium chloride.



4.2.1 Use diagram A to explain why sodium conducts electricity. (2)

It has delocalised electrons that are able to flow. ✓✓

Sea of electrons are able to move past the positive kernels

4.2.2 Solid sodium chloride does not conduct electricity. Explain this property of the salt with reference to the relevant sub-atomic particle/s. (2)

The ions are bonded via strong ionic bonds, ✓ these are rigid and don't allow

charge to flow. ✓

(electrostatic forces in crystal lattice mean charged are not free to move)

4.2.3 Name a state or phase in which sodium chloride is electrically conductive. (1)

Liquid (molten) |OR| aqueous solution ✓

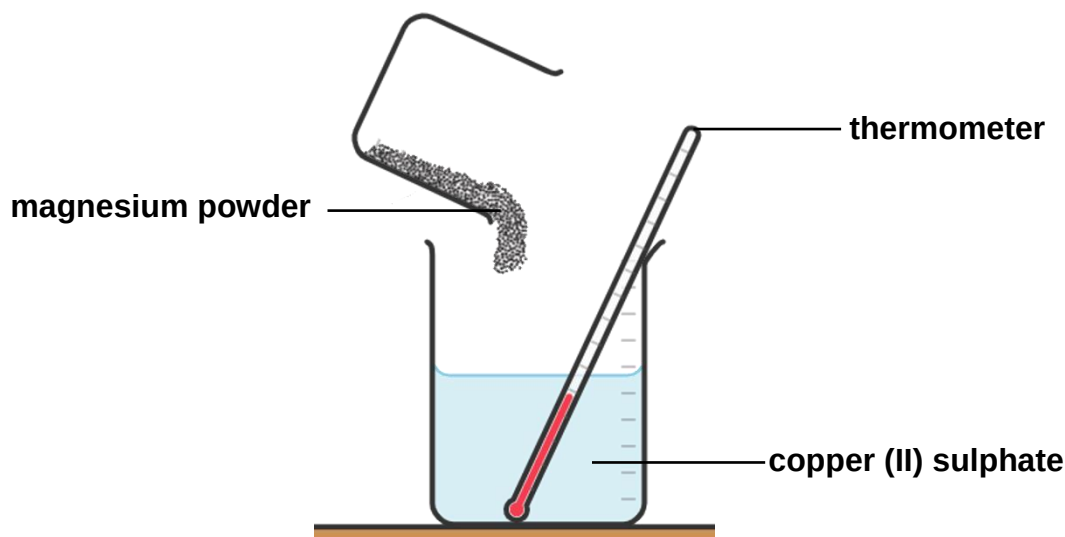
QUESTION 5

26 MARKS

A student investigated the relationship between the mass of magnesium powder and the change in temperature of a copper (II) sulphate solution.



They measured the temperature change when different masses of magnesium were added to a 50 ml copper (II) sulphate solution.



The student then repeated the experiment for each mass of magnesium three times and calculated an average. The table below summarizes his results:

Mass of magnesium powder (g)	Increase in temperature (°C)			
	Trial 1	Trial 2	Trial 3	Average
0,5	1,5	2,0	2,5	2,0
1,0	3,5	4,0	5,0	4,2
1,5	6,0	8,0	5,0	6,3
2,0	6,0	6,5	6,0	6,2
2,5	5,0	7,0	6,0	6,0

5.1 Define the term *exothermic reaction*. (2)

Reaction which converts chemical potential into thermal energy. ✓✓

5.2 Identify the dependant variable in the experiment. (2)

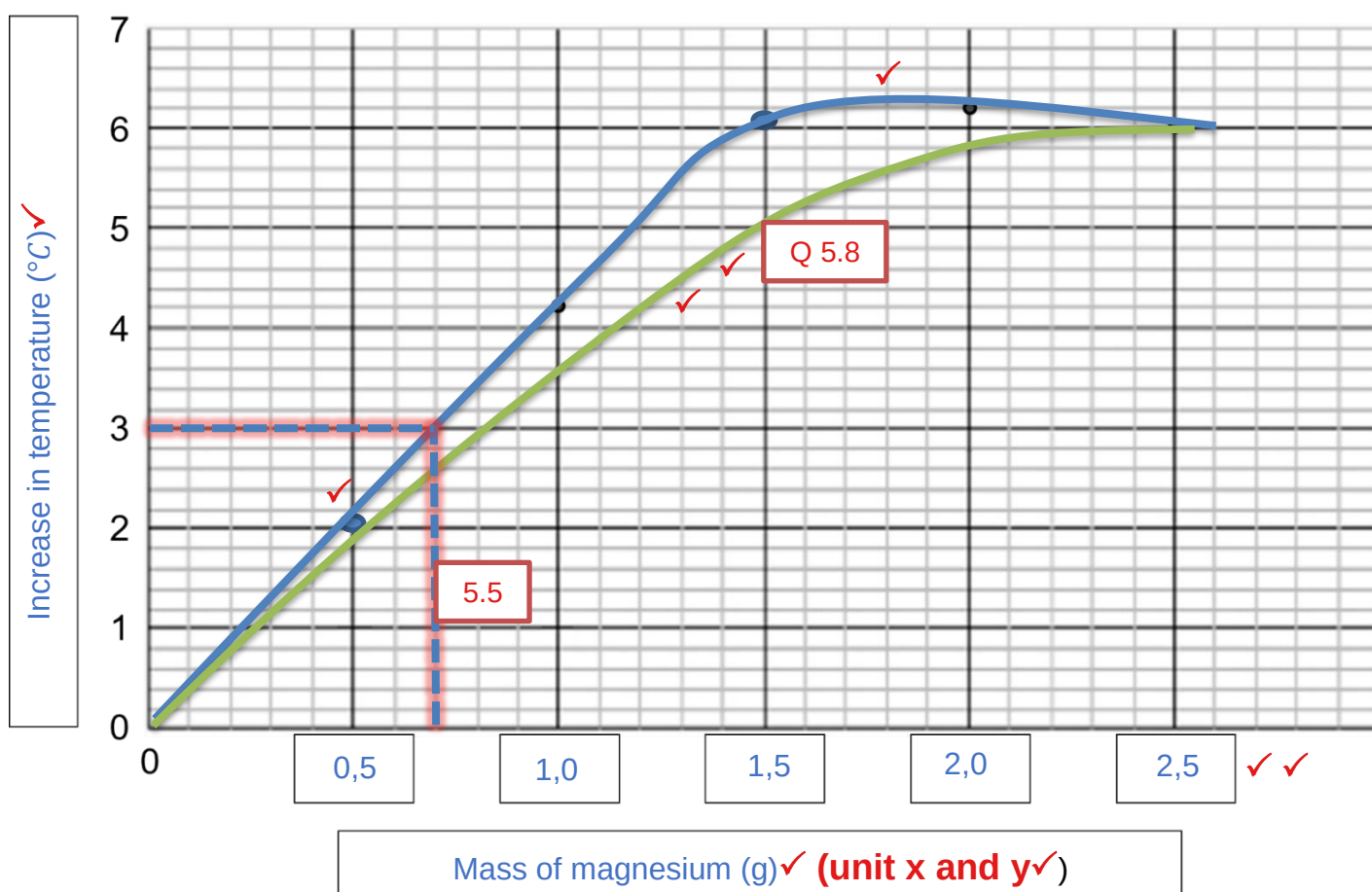
Temperature ✓ of solution ✓

- 5.3 The student examined their results and concluded that the 2,0 g of magnesium produced the most accurate results. Comment on the validity of his statement. (3)

It is precise but not accurate. ✓
- Precise as there is the least deviation from average ✓
- there is no control on which to test the results against to see how
accurate the value are. ✓

- 5.4 An incomplete graph of the results is shown below. Complete the graph by:

- 5.4.1 Providing the labels for the axes. (3)
 5.4.2 Providing a suitable scale for the x-axis. (1)
 5.4.3 Plotting the missing values. (2)
 5.4.4 Drawing a best fit curve for the data. (1)



- 5.5 Use your graph to find the mass of magnesium required to produce a temperature rise of 3°C. Show how you obtained your answer by drawing two clear dotted lines on the grid in the previous question. (2)

0,75 g ✓ (and ✓ for lines shown on graph)

- 5.6 Provide an explanation as to why the increase in temperature stays relatively constant for the last three mass readings (1,5 g, 2,0 g and 2,5 g). (2)

Reaction is complete, CuSO_4 is used up. ✓✓

[CuSO_4 is the limiting reagent]

- 5.7 Explain fully why the temperature rise increased as the mass of the magnesium powder increased from 0 to 1,5 g. (3)

There is a greater quantity of magnesium as mass increases. ✓

So more reactions (effective collisions) will occur per second. ✓

Releasing more energy (heat) per unit time. ✓

- 5.8 On the graph in QUESTION 5.4, show the effect on the temperature rises if the student were to repeat the experiment using the same masses of magnesium granules instead of magnesium powder. Label this new curve "Q5.8". (2)

- 5.9 Explain your answer to QUESTION 5.8. (3)

Using granules decreases the surface area able to react. ✓

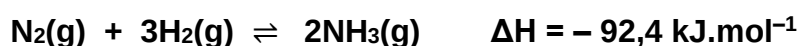
This means there are less reactions occurring per unit time. ✓

So the reaction rate decreases. ✓

QUESTION 6

17 MARKS

Ammonia is prepared in industry using the Haber process. The balanced chemical equation for the reaction is given below:



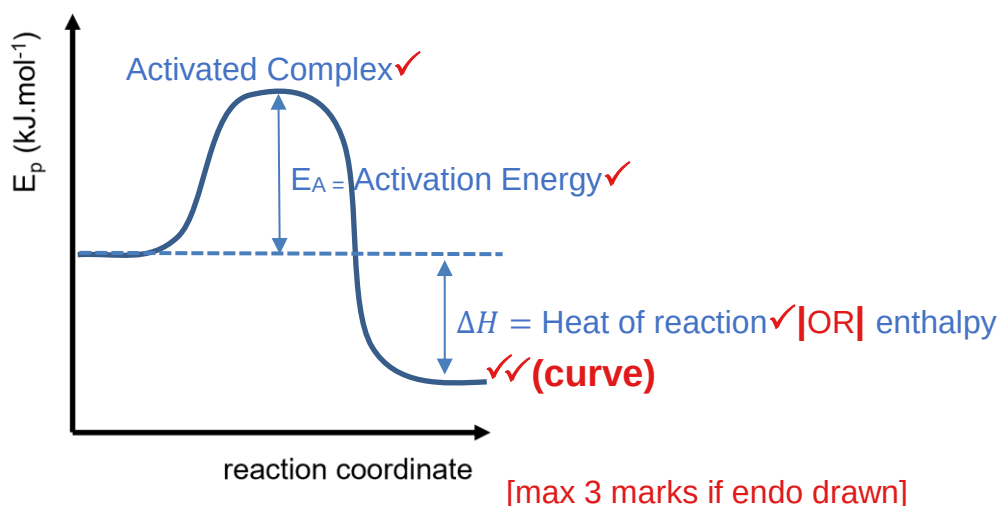
The activation energy for this reaction is $242,6 \text{ kJ.mol}^{-1}$.

- 6.1 Define the term *activation energy*. (2)

The minimum energy required to start a chemical reaction. ✓✓

| OR | The energy required to form the activated complex. ✓✓

- 6.2 On the axis provided below sketch the energy profile graph for this reaction. Include labels for activation energy, heat of reaction, and activated complex. (5)



- 6.3 Is energy **RELEASED** ✓ or **ABSORBED** when the bonds within ammonia are formed? Circle the correct answer. (1)

- 6.4 Calculate the activation energy required for the **reverse** reaction to proceed. (2)

$$\begin{aligned} \text{activation energy reverse} &= 242,6 - (-92,4) \\ \text{activation energy reverse} &= 335 \text{ kJ} \cdot \text{mol}^{-1} \checkmark \checkmark \end{aligned}$$

- 6.5 Define the term *catalyst*. (2)

A substance that increases the rate of the reaction ✓ but remains unchanged at the end of the reaction. ✓

- 6.6 A learner incorrectly states:
"A catalyst lowers the activation energy of the reaction which results in more collisions taking place per second."

Explain why this statement is incorrect. (2)

The actual activation energy of the reaction does not change, ✓
But it does provide an alternative path of lower activation energy. ✓

- 6.7 The following table shows two catalysts that can be used for this reaction and their corresponding activation energy of the reaction.

Catalyst	Activation Energy of the reaction. (kJ.mol ⁻¹)
Platinum Catalyst	70,6
Osmium catalyst	104,6

Which catalyst; platinum or osmium, will be most effective for this reaction?
Explain.

(3)

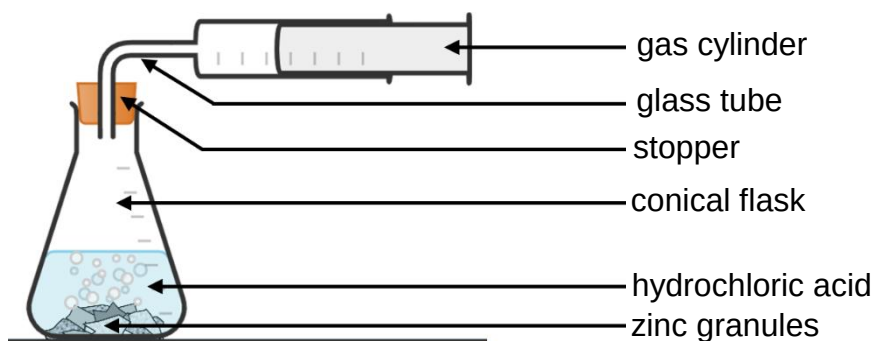
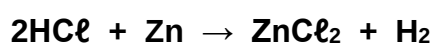
Pt. ✓
The lower the activation energy with a catalyst the higher the reaction rate will be. ✓
As the reaction will not require as much energy to reach the activated complex. ✓

QUESTION 7

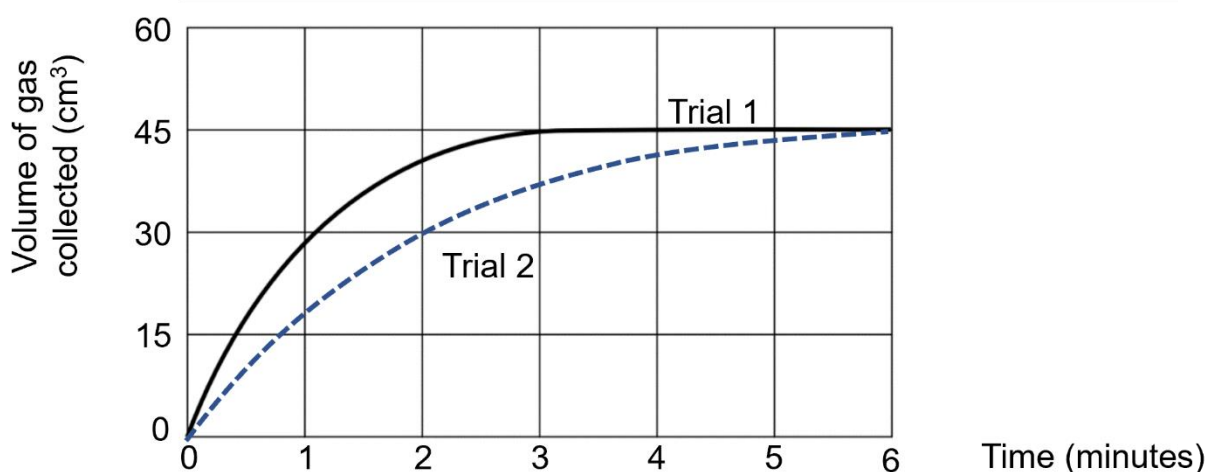
12 MARKS

A student wants to investigate how the concentration of hydrochloric acid influences the rate of reaction when HCl reacts with zinc granules. A conical flask is set up as shown in the diagram below. The student then records the volume of hydrogen gas collected.

The balanced chemical equation for the reaction is:



The following graph represents the students results. The solid line is the first trial of the experiment, and the dashed line is the second trial.



7.1 Define *reaction rate*. (2)

The change in concentration per unit time of either a reactant or product. ✓✓

7.2 Calculate the average rate of the reaction for Trial 1 between 0 and 3 minutes. (3)

$$\text{rate} = \frac{\text{measured quantity}}{\text{time}} \checkmark$$

$$\text{rate} = \frac{45}{3}$$

$$\text{rate} = 15 \text{ cm}^3 \cdot \text{minute}^{-1} \checkmark \checkmark$$

OR

$$\text{rate} = \frac{\text{measured quantity}}{\text{time}} \checkmark$$

$$\text{rate} = \frac{45}{180}$$

$$\text{rate} = 0.25 \text{ cm}^3 \cdot \text{s}^{-1} \checkmark \checkmark$$

7.3 Explain why the gradient of the graph for Trial 1 decreases over time. (2)

As the reactants are used up their concentration decreases, ✓ this means there

is less reactants available to react, decreasing reaction rate. ✓

7.4 Did the concentration of the hydrochloric acid **INCREASE** or **DECREASE** ✓ in Trial 2? Circle the correct answer. (1)

7.5 Explain your answer in 7.4 by referring to the collision theory. (4)

- A decrease in concentration means less particles per unit volume. ✓

- There will be less collisions per unit time. ✓

- There will be less effective collisions (with correct orientation and kinetic energy greater than activation energy) per unit time ✓

- The reaction rate decreases. ✓

QUESTION 8

13 MARKS

The fizz produced when an antacid dissolves in water is caused by the reaction between sodium hydrogen carbonate (NaHCO_3) and citric acid ($\text{H}_3\text{C}_6\text{H}_5\text{O}_7$). The balanced equation for the reaction is:



- 8.1 Write down the FORMULA of the substance that causes the fizz when the antacid dissolves in water. (1)

CO_2 ✓

- 8.2 Define *molar mass*. (2)

the mass in grams of one mole of that substance ✓✓

A certain antacid contains 3,36 g of NaHCO_3 , and 1,8 g of $\text{H}_3\text{C}_6\text{H}_5\text{O}_7$. The antacid is dissolved in 100 cm^3 distilled water in a beaker.

- 8.3 Calculate the number of moles of NaHCO_3 in the antacid. (3)

$$M(\text{NaHCO}_3) = 84 \text{ g} \cdot \text{mol}^{-1} \text{ ✓}$$

$$n = \frac{m}{M} \text{ ✓}$$

$$n = \frac{3,36}{84}$$

$$n = 0,04 \text{ mol} \text{ ✓}$$

- 8.4 Determine, using calculations, which substance is the limiting reagent. (4)

$$M_{\text{acid}} = 192 \text{ g} \cdot \text{mol}^{-1}$$

$$n = \frac{m}{M}$$

$$n = \frac{1,8}{192}$$

$$n = 0,009 \text{ mol} \text{ ✓}$$



$$0,04 : 0,009$$

$$0,04 \rightarrow 0,013 \text{ ✓}$$

$$0,027 \leftarrow 0,009$$

For all NaHCO_3 to react would require 0,013 moles of acid.

There is not enough acid therefore citric acid is the limiting reagent. ✓

8.5 Calculate the mass of the reactant in excess.

(3)

$$\text{excess} = \text{available} - \text{used}$$

$$\text{excess} = 0,04 - 0,027$$

$$\text{excess} = 0,013 \text{ mol} \checkmark$$

$$\therefore m = nM$$

$$m = (0,013)(84) \checkmark$$

$$m = 1,092 \text{ g} \checkmark$$

IF ROUNDED TO 2 DECIMAL PLACES

$$\therefore m = nM$$

$$m = (0,01)(84) \checkmark$$

$$m = 0.84 \text{ g} \checkmark$$

QUESTION 9

19 MARKS

- 9.1 Define the term *concentration*. (2)

The amount of solute per unit volume of solution. ✓✓

- 9.2 Eight (8) grams of $\text{Na}_2\text{S}_2\text{O}_3$ is dissolved in water to prepare 500 cm^3 of solution. Calculate the concentration of the $\text{Na}_2\text{S}_2\text{O}_3$ solution. (3)

$$M = 158 \text{ g} \cdot \text{mol}^{-1}$$

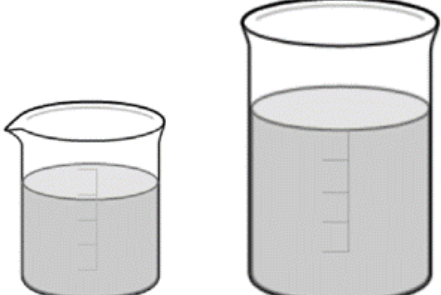
$$V = 500 \text{ cm}^3 = 0,5 \text{ dm}^3$$

$$c = \frac{m}{MV} \quad \checkmark$$

$$c = \frac{8}{(158)(0,5)} \quad \checkmark$$

$$c = 0,1 \text{ mol} \cdot \text{dm}^{-3} \quad \checkmark$$

- 9.3 You are given the following two aqueous solutions:

Solution A $c = 0,2 \text{ mol} \cdot \text{dm}^{-3}$ ammonium hydroxide $V = 0,25 \text{ dm}^3$		Solution B $c = 0,3 \text{ mol} \cdot \text{dm}^{-3}$ ammonium carbonate $V = 0,5 \text{ dm}^3$
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- 9.3.1 Give a reason why the above solutions are standard solutions. (1)

They have known concentrations. ✓

- 9.3.2 Define the term *solvent*. (2)

The substance in which another substance is dissolved, forming a solution. ✓✓

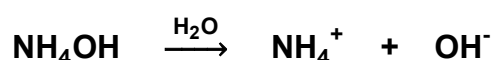
- 9.3.3 Identify the solvent used in the above solutions. (1)

water ✓

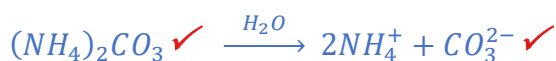
9.3.4 Give the method for making a standard solution using four key bullet points. (4)

• Measure out precise mass of solute. ✓
• Pour into beaker cleaning watch glass. ✓
• Add a volumetric flask and rinse beaker ✓
• Add last few drops to flask until filled to line. ✓

The balanced dissociation reaction for ammonium hydroxide is given as:



9.3.5 Write balanced chemical equation showing the dissociation of ammonium carbonate. (2)



9.3.6 Solution A is now mixed with Solution B in a new bigger beaker. Calculate the new concentration of the ammonium ions in the solution. (4)

<i>NH₄OH:</i> $n_1 = cV$ $n_1 = (0,2)(0,25)$ $n_1 = 0,05 \text{ mol} \checkmark$ <i>(NH₄)₂CO₃</i> $n_2 = cV$ $n_2 = (0,3)(0,5)$ $n_2 = 0,15 \text{ mol}$ $n_1 = 0,15 \times 2 = 0,3 \text{ mol} \checkmark$	<i>TOTAL MOLES:</i> $c = \frac{n_t}{V_t}$ $c = \frac{n_1 + n_2}{V_1 + V_2} \checkmark$ $c = \frac{0,05 + 0,3}{0,25 + 0,5}$ $c = \frac{0,35}{0,75}$ $c = 0,47 \text{ g} \cdot \text{mol}^{-1} \checkmark$ <i>c.o.e – 9.3.5</i> If ammonium carbonate mole not doubled: $c = 0,27 \text{ g} \cdot \text{mol}^{-1} \checkmark$
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QUESTION 10

20 MARKS

Study the unbalanced chemical equation of the reaction between sodium carbonate (Na_2CO_3) and hydrochloric acid (HCl) and answer the questions that follow.



10.1 Write a balanced chemical equation for the reaction with phase symbols. (3)



10.2 In a reaction, 15,9 g of sodium carbonate reacts completely with excess hydrochloric acid.

10.2.1 Calculate the initial number of moles of sodium carbonate. (3)

$$n = \frac{m}{M}$$

$$n = \frac{15,9}{106}$$

$$n = 0,15 \text{ mol}$$

10.2.2 Calculate the theoretical volume of CO_2 produced during this reaction. (4)



$$0,15 : 0,15$$

$$n = \frac{V}{V_m}$$

$$0,15 = \frac{V}{22,4}$$

$$V = 3,36 \text{ dm}^3$$

10.2.3 When the reaction was complete it was determined that only 15 g of NaCl was produced. Determine the percentage yield of the reaction. (5)

Theoretical:



$$\therefore 0,3 \text{ mol}$$

$$m = nM$$

$$m = (0,3)(58,5)$$

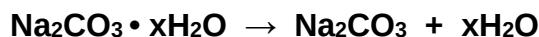
$$m = 17,55 \text{ g}$$

$$\% \text{yield} = \frac{\text{actual}}{\text{theoretical}} \times 100$$

$$\% \text{yield} = \frac{15}{17,55} \times 100$$

$$\text{yield} = 85,47 \%$$

- 10.3 14,2 g of a sample of hydrated sodium carbonate, $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$, was strongly heated until no further change in mass was recorded. On heating, all the water of crystallisation evaporated as follows:



Calculate the number of moles of water of crystallisation in the sodium carbonate sample, if 5,3 g of solid remained after strong heating. (5)

$$14.2 - 5.3 = 8.9 \text{ g of water} \checkmark$$



$$n = \frac{m}{M}$$

$$n = \frac{8.9}{18}$$

$$n = 0.49 \text{ mol} \checkmark$$



$$n = \frac{m}{M}$$

$$n = \frac{5.3}{106}$$

$$n = 0.05 \text{ mol} \checkmark$$

RATIO:

$$\frac{0.49}{0.05} = 9.8 \approx 10 \checkmark$$

$$\therefore \boxed{Na_2CO_3 \cdot 10 H_2O} \checkmark$$

END OF ASSESMENT

[illegible]